# Federated Learning Over Wireless Networks: Challenges and Solutions

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***Abstract*—Motivated by ever-increasing computational resources at edge devices and increasing privacy concerns, a new machine learning (ML) framework called federated learning (FL) has been proposed. FL enables user devices, such as mobile and Internet of Things (IoT) devices, to collaboratively train an ML model by only sending the model parameters instead of raw data. FL is considered the key enabling approach for privacy-preserving, distributed ML systems. However, FL requires frequent exchange of learned model updates between multiple user devices and the cloud/edge server, which introduces a significant communication overhead and hence imposes a major challenge in FL over wireless networks that are limited in communication resources. Moreover, FL consumes a considerable amount of energy in the process of transmitting learned model updates, which imposes another challenge in FL over wireless networks that usually include unplugged devices with limited battery resources. Besides, there are still other privacy issues in practical implementations of FL over wireless networks. In this survey, we discuss each of the mentioned challenges and their respective state-of-the-art proposed solutions in an in-depth manner. By illustrating the tradeoff between each of the solutions, we discuss the underlying effect of the wireless network on the performance of FL. Finally, by highlighting the gaps between research and practical implementations, we identify future research directions for engineering FL over wireless networks.**

***Index Terms*—Communication resources, federated learning (FL), power limitation, wireless networks.**

## I. Introduction

THE NUMBER of smart devices, such as mobile devices and Internet of Things (IoT) devices, is rapidly growing. Each of them generates and gathers its own data using its computational, communication, and sensor resources. These distributed data can be utilized to solve complex problems using machine learning (ML). Traditionally, devices send their raw data to a central coordinator (server) in which the coordinator trains the ML model based on the user devices' raw data. After the training process, the coordinator sends back the final model to user devices. However, with the increasing concern of privacy issues, this approach is not suitable for sensitive information since the coordinator can access each user's private information.

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To address the aforementioned challenges and exploit the abundance of user device data and their powerful computation resources, a distributed ML approach called federated learning (FL) has been proposed in [1]. In this approach, an FL system optimizes a global ML model by repeating the following process.

1) Each user device performs local computing with its data to minimize a global model sent by the FL coordinator. Then, they upload their local model updates to the FL coordinator for aggregation.
2) The FL coordinator aggregates the received local model from user devices and calculates an improved global model.
3) The coordinator then sends the new updated global model to user devices, and user devices use this new global model as the reference.

These steps create one global round (communication round) and multiple communication rounds are repeated until the desired accuracy is reached. This process has been illustrated in Fig. 1. FL allows the decoupling of data accusation and computation at the central unit [2] and enables devices to cooperatively train an ML model while ensuring their privacy.

The effectiveness of FL has been demonstrated in practical applications. For example, the federated averaging (FEDAVG) algorithm proposed in [1] has been applied to Google's Gboard [3] to improve next-word prediction. Further, applications in training ML models over high-sensitive data sets have also been proposed in [4]. In this study, Verma et al. utilized FL to build ML models of data across multiple agencies without transferring their data. Moreover, FL can be utilized *for wireless networks*. For instance, Niknam et al. [5] proposed several possible applications of FL in 5G networks in which FL can be utilized for spectrum management and edge caching. Furthermore, due to its privacy-preserving nature, FL is being treated as one of the vital solutions to perform distributed ML in 6G [6]. The potential applications of FL for 6G communication networks include ML-based energy optimization, ML-based performance networking, and new, emerging ML services such as intelligent transportation systems [6]. Further, the benefits of FL are also considered for aerial access networks [7]

With all the promising benefits, FL comes with its own specific challenges to tackle, specifically in the real-world practical implementation of FL over wireless networks. First, user devices in wireless networks usually include unplugged devices with limited battery resources. Therefore, their heterogeneous power limitation should be considered in FL. Second,

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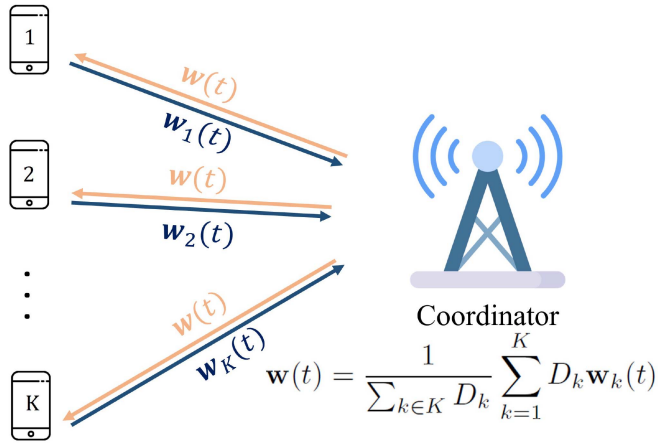


Fig. 1. Illustration of the FEDAVG process in wireless networks. In each global iteration $t$, each user device $k$ calculates its local model update $\mathbf{w}_k(t)$ based on the received global model $\mathbf{w}(t)$, and then send it to the coordinator. The coordinator then aggregates local models and updates the global model using weighted averaging of the local models for the next iteration. FEDAVG utilizes the size of the dataset of each user, i.e., $|D_k|$ where $D_k$ is the data set of user $k$. This process is done until the global model reaches a predefined level of accuracy.

wireless channel properties such as noise and interference, can cause inaccurate or unsuccessful transmission between user devices and coordinator, and therefore, affect the performance of FL. Third, the number of user devices and their ML models can be extremely large. Therefore, communication can still be a huge bottleneck in FL, and appropriate resource allocation and compression techniques should be utilized based on the physical layer implementation of FL over wireless networks. Finally, even though that user devices do not share raw information with the coordinator, adversaries can still infer sensitive information, if any, about individual devices based on the global model.

In this survey, we investigate the challenges of the implementation of FL over wireless networks. Although there are surveys on FL, the existing studies do not focus on the deployment challenges of FL over wireless networks, and more importantly, they did not cover the tradeoffs of the proposed solutions. For existing surveys on FL, the closest work to our survey is [8] in which Imteaj et al. discussed the overall challenges of deploying FL over IoT devices. However, the authors do not focus on the wireless communication aspect of deploying FL and briefly summarize some of the challenges. Liu et al. [6] provided potential applications and challenges of FL for 6G wireless networks. Similarly, Lim et al. [9] provided a survey on implementation challenges of FL over mobile-edge networks. However, they mainly focus on the challenges and applications of FL for edge networks. We select and summarize some of the closest surveys to our work in Table I. We refer you to [14] for a comprehensive list of surveys on FL. These surveys do not provide a detailed understanding of the challenges of deploying FL over wireless networks and their impact on the performance of FL. Further, they do not evaluate the proposed solutions for each challenge in the existing literature.

This motivates us to provide a survey to provide a detailed analysis of the challenges of implementing FL over wireless networks and their respective state-of-the-art solutions. Our main contributions are summarized as follows.

1) To highlight the effects of the challenges of implementing FL over wireless networks, we discuss each of the challenges by categorizing the state-of-the-art solutions for each challenge as shown in Fig. 2. By providing the fundamental tradeoffs between each solution and its performance, we illustrate the underlying effect of the wireless network on the performance of FL.
2) To fully capture implementation challenges, for each challenge, we provide a detailed overview of the implementation of the state-of-the-art solutions using the main system model.
3) We discuss future research directions by highlighting the gaps between research and practical implementation.
4) Finally, by summarizing the effects of wireless networks on the performance of FL, we illustrate that for maximum performance, in terms of bandwidth efficiency and learning time, both FL and the underlying wireless network should be specifically co-designed to complement each other.

The remainder of this article is organized as follows. Section II discusses the system model of FL. Section III discusses the power limitation challenges of implementing FL over wireless networks and the proposed solutions to overcome this challenge. Section IV discusses the effects of the wireless channel properties on the performance of FL and the proposed solutions to overcome this challenge. In Section V, we discuss the effects of limited available bandwidth as one of the main bottlenecks of implementing FL over wireless networks. Further, we discuss solutions to overcome this challenge based on the physical layer implementation of the network. In Section VI, we discuss the privacy issues in FL over wireless networks and the available solutions to this issue. Section VII discusses the future directions in FL over wireless networks, and Section VIII concludes this article.

## II. Federated Learning Model

In this section, we define a typical FL system based on the FEDAVG method as proposed in [1]. Let $\mathcal{K} = \{1, 2, \ldots, K\}$ denote the set of $K$ user devices in the wireless network. Each of these devices has a private dataset $D_k$ for $k \in \mathcal{K}$. At each global iteration $t$, each user device $k$ uses $D_k$ to train a local model $\mathbf{w}_k(t)$. Specifically, it finds the optimal parameters $\mathbf{w}_k^*(t)$ that minimizes the ML loss function $F(\mathbf{w}_k(t))$ as follows:

$$\mathbf{w}_k^*(t) = \min_{\mathbf{w}_k(t)\in\mathbb{R}^d} F(\mathbf{w}_k(t)) \tag{1}$$

where the loss function is selected based on the ML model.

The classical algorithm of FL is FEDAVG which is proposed in [1]. In this algorithm, as shown in Algorithm 1, for this local minimization, each user device runs multiple iterations (local steps) of stochastic gradient descent (SGD). Then, after a fixed number of local iterations, user devices send their locally updated model to the coordinator for aggregation. To do so, the coordinator uses an averaging algorithm to find the

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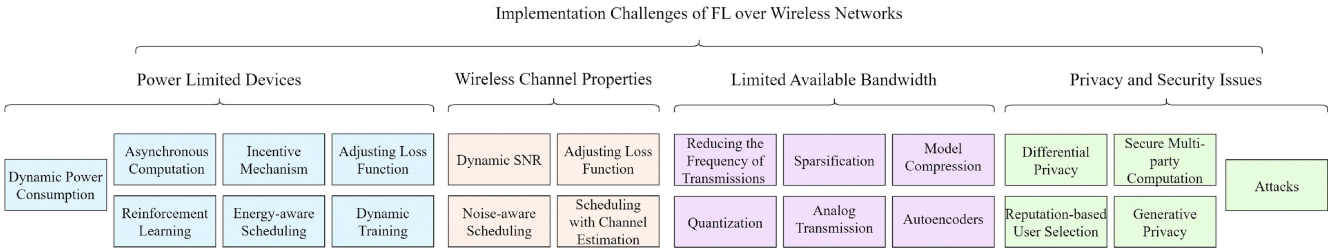


Fig. 2. Classification of research on challenges of FL over wireless networks.

TABLE I
EXISTING SELECTED SURVEYS ON FL

| Subject | Ref | Contribution |
|---|---|---|
| FL Deployment Challenges | [8] | Investigates the implementation challenges of training distributed machine learning models for IoT devices. |
| | [9] | Evaluates integration of FL and edge computing and discusses implementation challenges of FL over mobile edge networks |
| | [5] | Provides a short overview on the challenges and application of using FL for the next generation wireless communications |
| FL Architecture and Applications | [10] | Explores the potential of FL for enabling a wide range of IoT services and applications. |
| | [10] | Categorizes FL architecture based on its data partitioning, privacy mechanism, and machine learning model, and then analyzed each category. |
| | [6] | Provides potential applications and challenges of FL for 6G wireless networks. |
| | [11] | Review existing systems and infrastructure of FL systems, including system components, taxonomy, and design. |
| FL Security and Privacy | [12] | Identifies and examines security and privacy vulnerabilities in FL environments and provides insights into existing defense mechanisms. |
| | [13] | Examines security and privacy attacks to FL and evaluates solutions proposed in the literature to mitigate each attack. |

updated global model $\mathbf{w}(t)$ as follows:

$$\mathbf{w}(t+1) = \frac{1}{\sum_{k\in\mathcal{K}} |D_k|} \sum_{k=1}^{K} |D_k| \mathbf{w}_k(t). \tag{2}$$

The coordinator uses (2) to calculate the weighted average of all the locally updated models. The weights are selected based on the size of the data sets of each user device to make sure that the coordinator is calculating an unbiased estimate of global updated models.

The FL global iteration is performed until it reaches a predefined desirable level of accuracy. It is shown in [15] that FEDAVG algorithm converges with the rate $\mathcal{O}(1/T)$ on non-iid data sets for strongly convex and smooth loss functions. In this model, it is assumed that the underlying network can transmit both the updated local model and global model with perfect accuracy. As we discuss in Section IV, this assumption does not hold in most practical scenarios. Further, the FEDAVG algorithm (Algorithm 1) transmits the ML's model weights, $\mathbf{w}_k(t)$, to the coordinator and the coordinator aggregates these model weights for global aggregation. This method is called *model averaging*. Alternatively, it is possible to send gradient information of the SGD, $\mathbf{g}_k(t)$, instead of model weights to the coordinator, and the coordinator can aggregate gradients to calculate the updated global model. This method is called *gradient averaging*. In this survey, we refer to both these parameters as updated models. For more information on the differences between model averaging and gradient averaging we refer you to [16].

The FEDAVG algorithm discussed above utilizes the weighted averaging method with simple SGD to aggregate models. However, there are many other proposed aggregation methods in the literature, including FedProx [17], SCAFFOLD [18], FedAWS [19], FedAdam [20], and FedPer [21]. These methods have been proposed to address different challenges in deploying FL. For instance, FedAdam was proposed to improve the convergence rate of FL, and FedPer was proposed to address the data heterogeneity of FL users. In this survey, we are focusing on FEDAVG as our main aggregation method since it is the most widely analyzed method in both theory and experiments. However, we will discuss other aggregation methods that can be used to address the challenges of deploying FL over wireless networks.

The practical implementation of FL over wireless networks has unique challenges as compared to traditional distributed ML. In the following sections, we evaluate the effect of the underlying wireless network on the performance of FL. For this purpose, we will use learning time (convergence rate) as one of the key performance metrics of FL. In FL, learning time is defined as the time it takes for the complete training process to reach a predefined level of accuracy.

## III. POWER-LIMITED DEVICES

One of the key challenges of implementing FL over wireless networks is the limitation of energy in users' devices. Users' devices usually include unplugged mobile and IoT devices with limited batteries. Therefore, wireless devices must minimize their power consumption on both the local computation of the learning model and the transmission of the updated model with the coordinator. This transmission includes both downloading the updated global model from the coordinator and uploading the updated local model to the coordinator.

Given a predefined global level of accuracy of the training model, user devices need to allocate limited energy resources to both computation and transmission. However, this allocation has a direct impact on the learning time, which consists of the time of transmission and the time of local computation. Therefore, there is a tradeoff between computation energy, transmission energy, and learning time. Further, computation itself is constrained by device resources, such as maximum CPU frequency, and transmission is constrained by the device

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**Algorithm 1** FEDAVG Algorithm [1]

**Parameters:** $\eta$ = the learning rate of SGD, $\tau$ = the number of local iterations per communication round, $T$ = the total number of global iterations (which is selected based on the desired level of model accuracy)

**Initialize:** $\mathbf{w}(0) \in \mathbb{R}$
**for** $t = 0, 1, 2, ..., T-1$ **do**
  **for** each device $k \in \{1, 2, ..., K\}$ in parallel **do**
    Initialize $\mathbf{w}_k(t) = \mathbf{w}(t)$
    **for** $s = 1$ to $\tau$ **do**
      Sample $\mathbf{x}_s$ from $D_k$ uniformly at random, and update the local model as follows:

$$\mathbf{g}_k(t) = \nabla F(\mathbf{w}_k(t), \mathbf{x}_s)$$
$$\mathbf{w}_k(t) = \mathbf{w}_k(t) - \eta \mathbf{g}_k(t)$$

    **end for**
    Send the locally updated model $\mathbf{w}_k(t)$ to the coordinator
  **end for**
  The coordinator collects all of the updated local models and updates the global model as follows:
  $\mathbf{w}(t+1) = \frac{1}{\sum_{k \in \mathcal{K}} |D_k|} \sum_{k=1}^{K} |D_k| \mathbf{w}_k(t)$
**end for**

**Output:** $\mathbf{w}(T)$

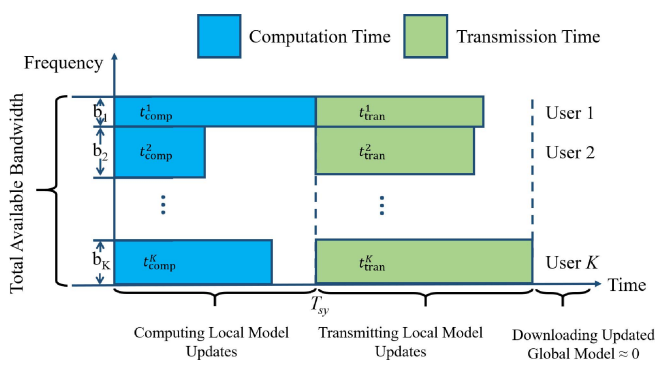


Fig. 3. Synchronous computation phase implementation of FL [22].

resources, such as maximum transmission rate, and network resources, such as overall available bandwidth.

A typical FL system has a synchronous computation phase where all devices have to finish solving their local problem with a local accuracy before a specific time, $T_{sy}$, when the communication phase begins as shown in Fig. 3. This type of implementation can utilize channel access methods like TDMA. The tradeoff between learning time versus the overall energy consumption of devices with synchronous implementation has been studied in [23]. In this scheme, unlike the traditional FL algorithm that runs the local iteration for a fixed number of iterations, user devices run the iterations until they reach a predefined level of local accuracy. In this synchronous implementation, for strongly convex loss functions, the general upper bound on the number of global iterations is shown to be [24] $([\mathcal{O}(\log[1/\epsilon])]/[1-\theta])$ which depends on both the local accuracy $\theta$ and global accuracy $\epsilon$. This means that for a predefined global accuracy, the number of transmissions to the coordinator is reversely correlated to the local accuracy which determines the number of local iterations.

In this section, we discuss the proposed solutions to address the power limitation of user devices.

### A. Dynamic Power Consumption

Tran et al. [23] formulated the minimization problem of the learning time and device energy consumption by using CPU energy cost for each iteration, the energy cost of transmission, and the delay of transmission. To minimize the overall energy cost and learning time at the same time, by decomposing the minimization into three convex subproblems, the closed-form solution is found for the CPU frequency, transmission rate, and local accuracy which determines the upper bound of the number of local iteration and global transmissions. In this study, the numerical results suggest that a high level of data heterogeneity has a negative impact on the total energy cost of FL.

### B. Energy-Aware Scheduling

Another solution to address the power limitation of user devices in FL is to use energy-aware scheduling and only allocate resources to devices with high computation capacity and good channel conditions [25]. However, since not all the available data is utilized in this case, the accuracy of the learning model decreases [25].

In a similar work, Hamdi et al. [13] studied the problem of joint user scheduling and energy management for minimizing the loss function in an FL network with a massive multiple-input–multiple-output (MIMO) coordinator. They show that maximizing the number of scheduled users enhances the convergence speed of the FL algorithm and that user scheduling strongly depends on the channel conditions and available energy. Further, they proposed an optimum algorithm for user scheduling and energy management. More recently, Chen et al. [26] followed this work and jointly optimized communication efficiency and resources for FL over wireless. In a separate work, Dinh et al. [27] minimized users' total overall training time in the FL setup by considering user devices' heterogeneous resources.

In another work, Beitollahi and Lu focused on the power limitation of the unplugged coordinator in FL systems that one of the unplugged users acts as the coordinator [28]. To address coordinators' power limitation due to uploading scheduling information, they proposed a stochastic-based multiframe scheduling approach where the coordinator sends the scheduling information at the beginning of each communication round and puts its communication device into sleep mode.

### C. Dynamic Training

To improve the energy efficiency of user devices, other approaches have also been explored. Li et al. [29] proposed a flexible compression control scheme to balance the tradeoff between local computation and transmission. To do so, SGD

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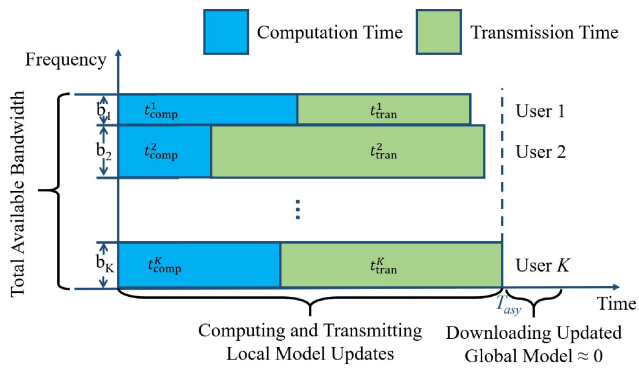


Fig. 4. Asynchronous computation phase implementation of FL [22].

with dynamic batch sizes and dynamic gradient sparsification is utilized. This approach enables each user device to flexibly adjust its computation energy cost by dynamically selecting the batch size for local computation and adjusting its communication energy cost by adjusting the sparsification level for compressing its updated local model. The numerical results for this convergence-guaranteed algorithm show that this flexible sparsification has better overall energy efficiency and learning time compared to when all user devices sparsify their updated models with the same level, especially with the increase in heterogeneity of user devices' communication capacity.

### D. Asynchronous Computation

Another method to address the power limitation of user devices is to enable users to finish the computation phase asynchronously. More specifically, in asynchronous computation implementation of FL over wireless networks, devices are not required to finish local training by a given time. However, all of the updated local models have to be transferred to the coordinator before a specific time, $T_{\text{asy}}$, to calculate and transmit the updated global model as shown in Fig. 4. The time constraints for the asynchronous computation phase in each communication round is $t^k_{\text{comp}} + t^k_{\text{tran}} \leq T_{\text{asy}}$ for all $k \in \mathcal{K}$, where $t^k_{\text{tran}}$ denotes the model transmission time of user $k$. The asynchronous computation phase can be utilized with channel access methods like frequency-division multiple access (FDMA) and orthogonal FDMA (OFDMA).

In this asynchronous implementation, since user devices can manage the available time of $T_{\text{asy}}$ for both computation and transmission we have a minimization problem with more freedom compared to synchronous implementation where the available time $T_{sy}$ is only used for computation. This opportunity can be utilized to minimize the overall energy cost of devices and the learning time. For instance, users with high energy-efficient CPUs can boost the frequency to run the iteration faster and send the data more slowly with less bandwidth to the coordinator. With this, they will enable low energy-efficient CPUs to lower their frequency and run the iteration longer and send the local model faster by using more bandwidth to make sure that it gets to the coordinator before $T_{\text{asy}}$ as shown in Fig. 4.

The tradeoff between learning time versus overall energy consumption of user devices has been studied in [22] for asynchronous implementation. Deriving the convergence rate of the asynchronous implementation, Yang et al. solved the minimization problem using an iterative algorithm, for which, at each iteration, closed-form solutions for computation resources (CPUs frequency) and transmission resources are derived. The result shows that using the flexibility of asynchronous implementation with adjustable bandwidth outperforms synchronous implementation in minimizing the overall energy cost and the learning time.

### E. Reinforcement Learning

To achieve the optimal energy-efficient resource allocation decision, the coordinator must access the energy, CPU, and transmission states of all user devices. However, these states can be uncertain or unknown. To address this issue and improve energy efficiency, Anh et al. [30] utilized the Deep $Q$-Learning technique to find the optimal energy management for user devices without any prior information on the network dynamics. To do so, the reward function based on the energy consumption, training latency, and the *informativeness* of user devices data is defined. In this approach, over time the algorithm observes the CPU state of user devices and accordingly decides the proper number of data units for local computation for each device such that the user devices finish the training process at the same time. Further, the algorithm observes the *usefulness* of each user device data and selects the proper number of data units for local computation for each device accordingly. The experimental results show that this approach is more efficient and has better learning time compared to the greedy scheme where all user devices' data is utilized.

### F. Incentive Mechanism

All the aforementioned studies assumed that all of the user devices participate in the FL unconditionally. However, this assumption does not hold in real-world scenarios because of the cost of resources used by the user devices such as power. To address this issue, Kang et al. [31] utilized the contract theory to design an incentive mechanism to encourage user devices with high-quality data to participate in FL. The quality parameter of local data is defined as the type of contract model, in which user devices with higher quality can receive more rewards. Numerical data shows that this scheme can attract user devices with high-quality data to improve the energy efficiency of FL and its learning time. Similarly, Khan et al. [32] proposed the Stackelberg game to motivate the participation of user devices.

### G. Adjusting Loss Functions

Typical FL setup requires tight synchronization across users since all the participating users run the same number of local iterations. To address this limitation and at the same time address the system heterogeneity of users, Sahu et al. proposed FedProx [33] in which they add a regularizer to the loss function to penalize the distance between the current local model and the current global model. This allows them to limit the divergence of each local model from the global model and

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allows users to run a different number of local iterations based on its power limitation in each global round.

### H. Review

Overall, using dynamic computation resources, such as adjustable CPU frequency, and dynamic transmission resources, such as adjustable upload rate or bandwidth, we can achieve less energy consumption and better learning time when channel state information (CSI) is available [22], [23]. Further, by efficiently selecting the local accuracy, we can adjust the ratio of local computation and the number of transmissions to achieve less energy consumption and better learning time [22], [23]. Alternatively, it is possible to consider energy-aware scheduling and only select devices with high computation capacity and good channel conditions to update to the global model at the cost of reducing the accuracy of the learning model [25].

We summarize the approaches for the challenge of power-limited devices along with references in Table II.

## IV. Wireless Channel Properties

Another key challenge of implementing FL over wireless networks is the effect of wireless channel properties and their time-varying conditions on the performance of FL. Adverse channel conditions can cause the transmission of the updated model either to be inaccurate or unsuccessful. For instance, inaccurate channel estimation, feedback quantization, delay in signal acquisition on a fading channel, etc. can make the updated model received in user devices or coordinator to be inaccurate [34] in practical implementations where noise is an indispensable part of the transmitted model. This can increase the learning time, especially for training models not robust to noise such as neural networks [35].

Moreover, wireless channels can make the transferred signal of the updated model to be undetectable and get lost. When an upload is unsuccessful, it might not be efficient to wait for the retransmission since the overall FL algorithm is iterative and can be sent in the next global transmissions. Further, because of the nature of training models, it can be even beneficial to randomly discard the device's updated models to prevent overfitting [36]. However, when the number of unsuccessful transmissions increases due to the decrease in signal-to-interference-plus-noise-ratio (SINR), the FL algorithm losses important information about its device's data sets and their computation. Therefore, to reach a predefined level of accuracy it must run more global iterations to transmit the lost information which eventually increases the learning time.

In this section, we discuss the effect of inaccurate and unsuccessful transmission on the performance of FL and the methods to address these issues.

### A. Dynamic Signal-to-Noise-Ratio

FL can compensate for some errors because of its iterative error–correction process. However, the main question is how much communication noise it can handle and what its impact is on the learning performance [37]. Wei and Shen [37] investigated the convergence of FL with strongly convex loss function under noisy communication and non-iid data sets with full or partial client participation. Analytical results reveal that, in order to maintain the same $\mathcal{O}(1/T)$ convergence rate of FedAvg with perfect communications, the uplink and downlink signal-to-noise-ratio (SNR), should be controlled such that they scale as $\mathcal{O}(t^2)$. This is because the time-varying learning rate, $\eta \sim \mathcal{O}(1/t)$, is large in the early stages and it can compensate for the noise, and when the learning rate gets smaller over time it is still bigger than the noise in each iteration since it is scaling slower than the noise. Therefore, the learning rate can always undo the effect of noise and make sure that the local model is converging to the optimal point.

The scaling SNR can be implemented using dynamic power transmission or diversity combining [37]. The numerical study in [37] demonstrates that fine-tuning an increasing power transmission (scaling SNR) outperforms equal SNR transmission under fixed overall available transmission power. It should be noted that even though the upper bound over convergence is unchanged in the scaling SNR method, the learning times increase because of the effect of noise in practice.

### B. Adjusting Loss Functions

Another approach to tackle the challenge of inaccurate transmissions is to adjust the loss functions. Training a loss function with noise is equal to adding a regularizer to the loss function [38]. To alleviate the effects of noise on FL, Ang et al. [34] proposed a regularizer and considered a new loss function for Gaussian noise for each device such that the new loss function flattens its region to resist the noise. This new loss function can achieve better learning time compared to using the original loss function in networks subject to noise [34].

### C. Noise-Aware Scheduling

To combat the effect of unsuccessful transmission, Yang et al. [39] proposed a noise-aware scheduling policy to select user devices with the best proportional noise condition at each communication round. The analytical results in [39] show that under high SINR and i.i.d. data sets, this policy outperforms in learning time compared to randomly selecting devices for uploading the updated local model and compared to the round-robin scheduling policy, where all the devices fairly upload their data in groups in each communication round. However, under low SINR, since not all the user devices use their data, round-robin and random scheduling outperform this policy.

### D. Scheduling With Channel Estimation

The scheduling policy in [39] and energy-efficient resource allocation in the previous section were derived under the assumption that accurate CSI is available at the coordinator. However, this assumption may not hold in the real-world practical implementation of FL over wireless networks. To address this issue, Wadu et al. [40] proposed a joint user scheduling and resource allocation policy under imperfect CSI. Specifically, Wadu et al. utilized Gaussian process regression to predict accurate channel information and then used this

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TABLE II
APPROACHES TO TACKLE THE CHALLENGE OF POWER-LIMITED DEVICES IN FL

| Approaches | Ref | Key Ideas | Tradeoffs |
|---|---|---|---|
| Dynamic Power Consumption | [23] | Adjusts the CPU cycle frequency, transmission rate, and local accuracy dynamically to minimize both power consumption and transmission time | Requires adjustable resources on user devices |
| Asynchronous Computation | [22] | Following [23], enables asynchronous transmission to further improve both power consumption and transmission time | Requires adjustable resources on user devices |
| Reinforcement Learning | [30] | Utilizes reinforcement learning to find the optimal energy management without any prior information on the network resources. | Lacks theoretical guarantee |
| Incentive Mechanism | [31] | Utilizes the contract theory to design an incentive mechanism to encourage user devices with high-quality data to participate in FL | Cannot utilize all the available information in the network |
| | [32] | Similar to [31] but uses the Stackelberg game to motivate the participation of user devices | Non-efficient resource utilization |
| Energy-aware Scheduling | [25] | Selects devices with high computation capacity and good channel condition to update the global model | Reduces the global model accuracy |
| | [13] | Formulates the problem of joint resource scheduling and energy management in FL network with MIMO coordinator | Requires instant information of user devices energy |
| Adjusting Loss Function | [33] | Adds a regularizer to the loss function to penalize the distance between the current local model and the current global model and allows each user to run a different number of local iterations | Increases learning time |
| Dynamic Training | [29] | Utilizes SGD with dynamic batch sizes and dynamic gradient sparsification to flexibly adjust computation energy cost and communication energy cost | Gradient sparsification is bounded by the learning rate |

estimation for scheduling decisions by utilizing the drift-plus-penalty technique from the Lyapunov optimization framework. The numerical results show that this approach leads to better learning time compared to the channel-aware scheduling policy in [39] for wireless networks with imperfect CSI.

### E. Review

Overall, both challenges of inaccurate and unsuccessful transmission should be tackled at the same time. For networks with inaccurate transmissions, adjusting the original loss function by adding a regularizer can redress the effect of statistically known properties, such as quantization error to a degree and improve the learning time [34]. Moreover, scaling transmission SNR by using methods like dynamic transmission power or diversity combining reduces the effect of inaccurate transmission in FL over wireless networks [37] with strongly convex loss functions. For networks with a high possibility of unsuccessful transmissions due to noise, interference, etc., channel-aware scheduling policies should be utilized to allocate resources to devices with better channel conditions to improve the learning time [39] since devices with better channel conditions can compensate for the unused information of other user devices.

We summarize the approaches for the challenge of wireless channel properties along with references in Table III.

## V. LIMITED AVAILABLE BANDWIDTH

FL requires the exchange of user models' parameters periodically, where the size of the model can be extremely large. Therefore, high communication cost is one of the main concerns of real-world implementation of FL, especially for bandwidth-limited user devices. For instance, the VGGNet architecture has approximately 138 million parameters (4264 Mb) [41]. On the other hand, user devices have limited communication resources, e.g., the maximum LTE uplink rate is 75 Mb/s in ideal, laboratory settings [42]. It would take around a minute to transmit VGGNet in each iteration of FL using the LTE network in ideal conditions. This fundamental communication bottleneck in FL is restricting the ML model complexity and user participation. Further, the communication capacity can vary across users and cause bottlenecks in FL's performance. Therefore, communication-efficient designs capable of addressing resource heterogeneity in user devices are one of the key requirements for the successful implementation of FL.

A wide variety of methods have been proposed to reduce the communication cost of an FL network. We categorize and summarize the substantial research on communication-efficient FL.

### A. Reducing the Frequency of Transmissions

These methods reduce the amount of communication between the server and users. For instance, in FEDAVG [1], each user performs multiple rounds of local SGD instead of only one to compute the local model before sending the local model to the server. At the cost of computation resources, it was observed that this method does not affect the convergence rate of the FL model significantly. However, in nonidentical data distribution across user devices, FedAVG reduces the convergence rate significantly [43]. Therefore, in practice, it is hard to choose the appropriate number of local rounds as it may vary a lot for different models and data distribution across users.

### B. Sparsification

These methods reduce the number of parameters for transmission by projecting the original vector to a sparse vector with few nonzero elements and sending the nonzero elements to the server. In gradient sparsification, some coordinates of the gradient vector are dropped based on certain criteria [44], [45], [46], which for instance can depend on the informativeness of the gradients. Top-$K$ gradient sparsification [47] is one of the notable trends in which only the top $K$ values of each gradient of the clients are sent to the server. However, this method uses the same $K$ for all user devices, and

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TABLE III
APPROACHES TO TACKLE THE CHALLENGE OF WIRELESS CHANNEL PROPERTIES IN FL

| Approaches | Ref | Key Ideas | Tradeoffs |
|---|---|---|---|
| Dynamic SNR | [37] | Utilizes dynamic power transmission or diversity combining to scale both uplink and downlink SNR as $\mathcal{O}(t^2)$ to maintain the same convergence rate of FL with perfect communications | Requires instant information of noise |
| Noise-aware Scheduling | [22] | Schedules user devices with best proportional noise condition | Decreases the global model accuracy |
| Adjusting Loss Functions | [34] | Adds a regularizer to the model's loss function to approximate the original loss function by flattening its region to resist the noise | The model could stuck in local optimum |
| Scheduling with Channel Estimation | [40] | Utilizes Gaussian process regression to predict accurate channel information and then used this estimation for scheduling decisions by utilizing the drift-plus-penalty technique from the Lyapunov optimization framework | Susceptible to prediction errors |

therefore, it is bottlenecked by the user with the worst communication capacity. Further, this method uses gradient instead of model parameters for transmission, and therefore, each client can only run one local round of SGD before uploading the local model which increases the communication cost. Moreover, this vanilla sparsification fails to capture and utilize redundant, similar information across users. To address this issue, Beitollahi et al. [48] proposed dynamic sparsification to allow different values of $k$ for each user.

### C. Quantization

Quantization methods reduce the entropy of the weight updates by mapping all updates to a reduced set of values. For instance, SignSGD was proposed in [49] to quantize each gradient update to its binary sign and reduce the bit size per update or to send two bits of information per value as in TernGrad [50]. To address resource heterogeneity across users, a multiple access channel (MAC) aware gradient quantization scheme for FL has been proposed in [44], where user devices quantize their local gradient based on individual quantization budgets.

To extend quantization and reduce the unused quantized space, only the region occupied by the model parameters was dynamically quantized in [44] and [51] by using the maximum and minimum values. Analytical analysis in [44] shows users with a higher dynamic range (difference between maximum and minimum values) must be assigned a higher quantization budget, meaning that these users should send more bits in that iteration. This dynamic range represents the informativeness of the gradient of each user in some aspects. This contrasts with the traditional approach where the coordinator assigns equal quantization levels to all users. One important constraint is that the underlying wireless system should be able to support this quantization budget, and the quantization budget should be selected based on the capacity region of the MACs. For this reason, Chang and Tandon [44] formulated the problem of finding the optimal quantization level of each user at each iteration subject to the capacity region of MAC. This problem is an NP-hard problem, however suboptimal solutions by relaxing the constraints, such as full client participation can convert this problem to convex optimization. The result shows that this quantization outperforms uniform quantizations. However, dynamic quantization methods require additional communication where they need to send the maximum and minimum values of the gradients with full resolution to the coordinator.

Even though quantization methods have theoretical convergence guarantees for some FL setups [52], their performance does not match the sparsification methods. In practice, quantization methods and sparsification methods are used together. Amiri and Gündüz [53] proposed a sparsification and quantization scheme for FL called "D-DSGD" where they used both quantization and sparsification to reduce the communication cost. Further, they utilized error accumulation, where they propagate the current compression error to the next iterations. Basically, the idea is that the accumulation of small errors over iterations creates a fair resource allocation for all the entries of gradients. It has been shown that the FL convergence time can be improved by propagating the current compression error to the next iterations [54], [55], [56].

### D. Analog Communication

With analog communications, the local gradients are scaled and directly transmitted over the wireless channel. Since the coordinator is not interested in the individual gradients, but only in their average, the superposition nature of wireless channels can be utilized. Amiri and Gündüz [53] proposed A-DSGD, an "over-the-air" analog scheme, which does not employ any digital code either for compression or channel coding, and instead all the devices transmit their gradient estimates simultaneously in an uncoded manner. The underlying wireless MAC provides the sum of gradients for the coordinator. The main challenge of analog transmission is to align the power transmission of users from different distances to make sure the received signal at the coordinator represents the sum of the gradients. For reducing the size of the gradient in analog transmission Amiri et al. utilized error-accumulating compression using sparsification to reduce the size of the user's gradient to the size of the available bandwidth. The numerical results show that this analog scheme outperforms digital transmission schemes like D-DSGD, QuantizedSGD, and SignSGD in terms of learning time. Further, the numerical results show that this analog scheme is more attractive to low-power devices compared to digital schemes since it allows them to align their limited transmission power to dominate the noise [53]. Amiri and Gündüz [57] extended this work for wireless fading channels.

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The bottleneck of this setup is the devices with the longest propagation distance to the coordinator. This bottleneck happens when these devices maximize their allocated transmission power, and to align the transmission power, nearer devices have to adjust their power accordingly by using less power, and therefore, reducing their SNR which causes their signal to get lost under noise. It is possible to discard user devices with the longest distance to the coordinator in order to improve the SNR and accuracy of received updated models. However, discarding those users and their data can result in a less accurate model. For this reason, Zhu et al. [58] studied the tradeoff between SNR and the fraction of data exploited in learning in cellular networks. Recently, Wang et al. [59] considered a multicell wireless network where each cell performs a different FL task using analog communication. To address the intercell interference in both the downlink and uplink model/gradient transmissions, they proposed a cooperative multicell FL optimization framework to achieve efficient interference management for downlink and uplink transmission design.

### E. Model Compression

This approach changes the user's model to a less complex model to reduce the number of model parameters required to be transmitted for FL. In this regard, knowledge distillation is proposed in [60] in which the original teacher model is transferred to a smaller student model, and therefore, this smaller model could be utilized for FL at the cost of losing the encapsulated information in the main model. In federated dropout [61], the original models are replaced with submodels by zeroing out a fixed number of activation nodes at each fully connected layer. These model compression methods usually cannot be generalized since rebuilding a smaller model varies for different model architectures. Another approach is to combine split learning methods with FL to split a large model across multiple devices and perform the training method on each portion as proposed in [62]. We refer you to [63] for more information regarding model compression methods.

### F. Autoencoder

This approach dynamically encodes the values of model parameters of each user such that they can be decoded on the server. In this method, instead of zeroing out the model parameters like sparsification, we use autoencoders to construct both encoder and decoder to allow the server to reconstruct the entire model parameters. In this regard, Beitollahi and Lu [64] proposed FL with autoencoder compression (FLAC) that trains an autoencoder to encode and decode users' models at the server in the *Training State*, and then, sends the autoencoder to user devices for compressing local models for future iterations during the *Compression State*. To guarantee the convergence of the FL, FLAC dynamically controls the autoencoder error by switching between the training state and compression state to adjust its autoencoder and its compression rate based on the error tolerance of the FL system. They theoretically prove its convergence in some settings. The main issue with this approach is that it requires additional computation on the server.

### G. Review

Overall, compression techniques like sparsification and quantization are necessary for the practical implementation of FL over wireless networks. Even though the results suggest that analog schemes can be superior in terms of bandwidth efficiency, analog schemes require tight synchronization for the alignment of the transmissions. Moreover, they are not backward compatible, meaning that they cannot be easily implemented by the existing digital schemes, and also error control codes cannot be used in these schemes [44]. Finally, since the analog scheme utilizes a synchronized computation phase it is more prone to slow users. However, the mentioned studies in this section are a framework where further improvements, such as dynamic power allocation, dynamic sparsification level, and dynamic user selection can be utilized.

### H. Other Related Works

Amiria et al. [65] studied the scheduling policies for D-DSGD. The numerical results show that allocating channel resources based on both the channel conditions and the significance of local model updates has better learning time compared to scheduling policies based only on either of these two metrics individually. Amiria et al. used the $l_2$-norm of the change of the locally updated models in each global iteration for determining the significance of the locally updated models. Further, it is shown that when data is identically distributed, selecting only a single device at each global round provides better performance than sending multiple updated models with less accuracy. However, when the data distribution is non-i.i.d, scheduling multiple user devices provides better performance than scheduling only a single device. Similarly, Yang et al. [66] used both the channel quality and metric of *age of update* to schedule user devices based on the staleness of the updated models, i.e., user devices which did not send their updated local models in a long time get priorities in the scheduling. The numerical results show that this low-complex approach can improve the FL accuracy.

In order to address the non-i.i.d. data sets and energy limitation of user devices in analog aggregation, Sun et al. [67] proposed an online energy-aware dynamic user scheduling policy to maximize the average number of user devices for gradient update at each global iteration under energy constraints. In this work, to improve the FL performance under non-i.i.d. data sets, *data redundancy* is utilized by data exchange between user devices. The numerical result shows that data redundancy significantly improves the overall FL accuracy when data is non-i.i.d.

In the A-DSGD analog scheme, it was assumed that all user devices have CSI and therefore can cancel the fading effect when transmitting the locally updated model. Amiri et al. [68] proposed an analog scheme for when CSI is only available at the coordinator. In this scheme, Amiri et al. utilized multiple

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antennas at the coordinator and designed a linear receive beamformer at the coordinator to mitigate the fading effect, and then aligned the updated models. The results show that increasing the number of antennas at the coordinator provides a better estimate of the average of the gradients and improves FL accuracy. Jeon et al. [69] extended this work by improving the estimation performance by utilizing a compressive sensing algorithm for the coordinator to find the estimate of the transmitted signal by exploiting its sparsity in massive MIMO communication systems.

In the over-the-air analog scheme, since the superposition nature of the wireless network is utilized in an encoded manner, the coordinator receives the aggregated global model plus the noise of the channel. To address the effect of this noise, Sery et al. [70] proposed a convergent over-the-air FL (COTAF) algorithm in which they introduced time-varying scaling and precoding to the transmitted signals and used the fact that the expected difference in each set of SGD iterations is expected to gradually decrease in the training process. Further using *aggregation mapping* at the server side, this proposed system mitigates the noise induced by the channel over time.

To address the communication cost, Chen et al. [71] proposed a multihop framework that each intermediate router performs model aggregation before sending the data to the next hop to reduce the outgoing data traffic. This allows the network to aggregate more models under limited communication resources.

We summarize the approaches for the challenge of limited available bandwidth along with references in Table IV.

## VI. Privacy Issues

One prominent advantage of FL is that it enables user devices to train a learning model without exchanging the raw data of user devices. However, private information can still be divulged to some extent by analyzing the differences between the related uploaded models [73], [74]. For instance, Shokri et al. [75] showed that with access to only the trained ML model of the private information of a particular hospital, attackers can infer whether an individual was a patient at that hospital.

In general, honest but curious coordinators, dishonest users, and eavesdroppers in the wireless network can still exploit the FL system to extract information about user devices information. To address this issue, privacy-preserving techniques, such as differential privacy (DP) and secure multiparty computation (SMC) have been utilized in the framework as described next.

### A. Differential Privacy

DP is a mathematical framework wherein an algorithm may be described as differentially private if and only if the inclusion of a single instance in the training data set causes only statistically insignificant changes to the algorithm output [76]. DP puts a theoretical limit on the influence of a single individual.

To achieve DP, each user adds artificial noise to its updated model and sends only this randomized version to the coordinator in order to protect both the user devices and the coordinator against private information leakage. This artificial noise is proportional to the sensitivity of the output, where sensitivity measures the maximum change of the output due to the inclusion of a single data instance. More specifically, one popular way to achieve DP is to add Gaussian noise to the locally updated models [73].

Similarly, since the broadcast from the coordinator is also susceptible to adversaries, Shokri et al. [75] found the properties of the artificial Gaussian noise to be added in the downlink by the coordinator. Deriving the convergence bound of the FL, it is shown that there is a tradeoff between the convergence performance and privacy protection levels, meaning that by increasing $\epsilon$, i.e., relaxing the privacy protection level, the number of global iterations to achieve a predefined level of accuracy using the above algorithm will decrease.

Seif et al. [77] showed that analog communication schemes can also provide better DP in conjunction with bandwidth efficiency. For this reason, the superposition nature of the wireless medium is utilized to provide DP. It is shown that in the analog scheme the privacy leakage per user, $\epsilon$, scales as $\mathcal{O}(1/\sqrt{K})$ compared to orthogonal transmission in which $\epsilon$ scales as a constant. Further, it is possible to utilize the channel noise to provide DP in this analog scheme. For this reason, Liu and Simeone [78] utilized dynamic power control to adjust the transmission signal to noise. This approach aims to minimize the learning performance under power and privacy constraints.

### B. Secure Multiparty Computation

SMC methods enable multiple parties to jointly compute a function over their inputs without revealing those individual inputs [79]. SMC can be achieved using different techniques such as homomorphic encryption. An additively homomorphic encryption scheme is one wherein $Enc(m_1) \circ Enc(m_2) = Enc(m_1 + m_2)$ for some predefined function $\circ$. One such homomorphic scheme is the Paillier cryptosystem [80]. Additively homomorphic encryption can be utilized in the coordinator to calculate the average of the updated models without knowing the individual updated local model of each device. However, this encryption adds communication overheads. Another disadvantage of this approach is that after the calculation, user devices can decrypt the broadcasted output of the coordinator and access the raw output of the global model, and therefore infer information about other user devices.

### C. Hybrid Approach

To overcome the challenges of low accuracy of the DP and vulnerability of SMC to interference over outputs, Truex et al. [76] proposed a hybrid approach to balance these tradeoffs. Combining the DP and SMC, Truex et al. reduced the growth of noise injection while maintaining a predefined level of trust. It is shown that when applying the SMC, user devices can add a fraction of the noise to ensure the same DP guarantee while reducing the added artificial noise. This

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TABLE IV
APPROACHES TO TACKLE THE CHALLENGE OF LIMITED AVAILABLE BANDWIDTH

| Approaches | Ref | Key Ideas | Shortcomings |
|---|---|---|---|
| Reducing the frequency of transmissions | [72] | Utilizes multiple numbers of local steps of SGD to reduce communication cost | Decrease convergence rate in non-iid data distribution settings |
| Sparsification | [48], [44], [47] | Reduces the number of parameters for transmission by projecting the original vector to a sparse vector | Decreases convergence rate in non-iid data distribution settings |
| Autoencoders | [64] | Reduces the number of parameters for transmission by encoding model parameters and decoding at the server using autoencoders | Requires additional computation for training the autoencoder |
| Quantization | [50], [49] | Reduces the entropy of the weight updates by mapping all updates to a reduced set of values | Cannot address a large number of model weight parameters |
| | [44] | Following the works in [50] but user devices quantize their local gradient based on individual quantization budgets and channel condition | The convergence guarantee issues |
| | [53] | Utilizes both sparsification and quantization scheme for FL over MAC in wireless networks | Further investigation is required to know the effect of sparsification on the convergence rate |
| | [65] | Following [53], utilizes $l_2$-norm of the change of the locally updated model as a metric for the significance of local models to schedule user devices | Decreases the convergence time but convergence to the optimal point is unknown |
| Analog Transmission | [53] | Proposes an over-the-air analog scheme that all the devices transmit their sparsified and compressed gradient estimates simultaneously in an uncoded manner. The underlying wireless MAC provides the sum of gradients for the coordinator | Requires CSI and tight synchronization for the alignment of transmissions and also error control codes cannot be utilized |
| | [68] | Following [53], utilizes multiple antennas at the coordinator and designed a linear receive beamformer at the coordinator to mitigate the fading effect for when CSI is only available at the coordinator | |
| | [67] | Follwing over-the-air method, utilizes *data redundancy* by data exchange between user devices to address the non-i.i.d. datasets and energy limitations | |
| | [70] | Follwing over-the-air method, utilizes time-varying scaling and precoding to the transmitted signals to mitigate the effect of aggregated noise in the coordinator | |
| Model Compression | [72] | Changes the user's model to a less complex model to reduce the number of model parameters | Requires design choices for each specific model |

approach however cannot deal with user devices dropping out during the FL process [81].

### D. Reputation-Based Scheduling

The aforementioned techniques can only preserve the privacy of user devices to a degree during the FL training process, and still, user devices can mislead a global model by intentional or unintentional behaviors. User devices may upload unreliable or low-quality data to the coordinator. For this reason, Kang et al. [82] proposed a *reliable* FL in which they defined a metric for the reputation to select reliable and trusted user devices for the learning task. For this reputation metric, in each FL training, the user devices obtain rewards according to their data contribution and model training behaviors. This reputation is managed in a decentralized manner by utilizing a consortium blockchain deployed at each coordinator. The numerical results suggest that this approach improves the accuracy of learning models subject to attacks by eliminating the impact of unreliable workers.

### E. Generative Privacy

In the framework of FL, it is hard to verify user devices updated local models when they do not release information. To address this issue, Triastcyn and Faltings [72] proposed federated generative privacy. In this approach, user devices produce artificial data that can replace their real data. Since user devices may have insufficient data to train this artificial data locally, the FL framework is used to train generative adversarial networks collaboratively among themselves. This model produces samples from cross-user distribution and does not use information about specific users. This approach enables user devices to release their individual artificial data without compromising their privacy. These data can be used for verification, labeling, and future ML training.

### F. Security Issues

Ensuring privacy is one of the main objectives of FL, and therefore, to implement FL over wireless networks a reliable hybrid method should be utilized to ensure the privacy of each user device from both outsiders and insiders. However, it should be noted that these methods only provide a limited degree of privacy, and more importantly, FL is still susceptible to other security issues, such as poisoning attacks [83], backdoor attacks [84], and channel attacks [85]. Further, these approaches provide only data-related protection. The nature of wireless access medium is susceptible to communication attacks, such as jamming [86] and Denial-of-Service (DoS) [87]. We refer you to [12] for a detailed survey on security issues in FL.

We summarize the approaches for the challenge of privacy in FL along with references in Table V.

## VII. FUTURE RESEARCH DIRECTIONS

For the practical implementation of FL over wireless networks there are still gaps between real scenarios and the available research. Further, the challenges discussed in this survey are only a part of the challenges in the implementation

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TABLE V
APPROACHES TO TACKLE THE CHALLENGE OF PRIVACY IN FL

| Approaches | Ref | Key Ideas | Shortcomings |
|---|---|---|---|
| Differential Privacy | [75], [73] | Adds Gaussian noise to the locally updated models to achieve DP, | Decreases the learning time |
| | [78], [77] | Following [73], utilizes the channel noise in an analog scheme to provide differential privacy | |
| Differential Privacy and SMC | [76] | Combining the DP and SMC, reduces the growth of noise injection while maintaining a predefined level of trust | Cannot deal with users dropping out |
| Generative Privacy | [72] | Utilizes generative adversarial networks to produce artificial data locally to replace real data | Convergence and model accuracy issues |
| Reputation-based User Selection | [82] | Defines a metric for the reputation to select only reliable and trusted user devices | Convergence issues |

of FL over wireless networks at scale. We discuss some of these gaps and challenges in this section.

1) *Physical Layer Implementation:* The main requirement for the practical implementation of FL over wireless networks is the design of the physical layer. The approaches discussed in Section V, including [44] and [53] are a framework that enables this implementation. However, much more consideration is required to fully utilize these approaches in practical scenarios. For instance, the analog transmission scheme shows great promise in terms of bandwidth efficiency, but the effect of different channel conditions, such as multipath fading on this scheme should be investigated. Further, the effect of dropouts, slow users, and synchronization issues should be considered in this scheme. Further, the huge size of model parameters and user devices is the main challenge in the digital implementation of FL. New approaches both in model compression and resource allocation are required in digital communication schemes to close the huge gap between the performance of FL in the analog scheme and the digital scheme. This allows FL to be more efficiently implemented in the existing digital communication systems.
2) *Inaccurate and Unsuccessful Transmissions:* As discussed in Section IV, even though a small amount of inaccurate transmission and unsuccessful transmission can be useful in the training of an ML model, the overall effect of these issues on the performance of FL should be considered. Wei and Shen [37] showed how much inaccuracy can be tolerated without affecting the performance of FL, and this degree of freedom allowed them to improve the performance of FL in the same communication conditions. Therefore, the effect of unsuccessful transmissions or dropouts on the performance of FL should also be investigated to fully understand the real-world conditions of wireless networks which are unreliable and subject to interference. The results can enable us to improve the performance of FL by adjusting or designing communication systems. For instance, an important question to consider is whether inaccurate model transmissions or unsuccessful transmissions are preferable. How many inaccurate transmission or unsuccessful transmissions can the network tolerate without affecting the performance of FL?
3) *Asynchronous FL and Decentralized FL:* The FL model that we described in Section II and analyze throughout the rest of this survey is both centralized and synchronous, i.e., there was a coordinator to aggregate updated models, and the coordinator would wait for the local models to aggregate their data and then broadcast this information to user devices. However, it is possible to implement FL asynchronously in which user devices can send their local models without synchronization and the coordinator updates its global model whenever it receives an updated model from a user device. This type of implementation is more robust to slow user devices and can achieve a better learning rate [88]. However, the real-practical implementation of this approach specifically in wireless networks requires further investigation. Moreover, FL can be implemented in a decentralized manner without requiring a coordinator [89]. For example, mobile devices, IoT devices, or even moving vehicles can collaboratively train an ML model without requiring a coordinator such as an access modem. In this approach user devices can fully utilize node-to-node bandwidth [89] and improve the performance of FL. Future exploration of these new approaches and their convergence analysis can enable further improvement in the application of FL. For instance, one important question is how can we utilize the resources of the fast users without biasing the final model toward the faster user's data.
4) *Heterogeneous Federated Learning Over Wireless Networks:* Throughout this survey, we focused mainly on homogeneous models where models of users are the same. However, users' resources are not the same across devices. To address this issue and allow utilizing different sizes or architectures of ML models many approaches have been proposed in which Knowledge distillation [60] is the core component where a model can transfer its knowledge to another model. However, it is still unclear the best strategy for the network to enable and accommodate knowledge transfer. For instance, recently, Wang et al. [90] proposed utilizing a digital twin (DT) to train a large model on the server and distill the knowledge to a small model on user devices.
5) *Scalability:* To allow FL-trained models to compete with large language models such as GPT-4 [91] that were trained on millions of data samples, we need to utilize all the privacy-constrained data of million user devices. Currently, FL experiments are conducted at most with a few hundred devices. The practical scalability of the

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most basic FL method (FEDAVG) to millions of devices across the globe is still an unresolved question. All the challenges discussed in this survey should be addressed for this setting.

## VIII. CONCLUSION

In this article, we present a survey on the challenges of implementing FL over wireless networks and their respective solutions. These challenges include power-limited devices, wireless network characteristics, limited available bandwidth, and privacy issues. By providing an in-depth analysis of the state-of-the-art approaches for these challenges, we illustrate the fundamental tradeoffs in FL implementation. This analysis illustrated that for maximizing the performance of FL over wireless networks, the FL algorithm and the underlying network should be specifically designed to complement each other. Finally, by highlighting the gaps between research and practical implementation, we discussed future research directions.

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